

MODULAR KOSZUL DUALITY FOR CHIRAL ALGEBRAS ON ALGEBRAIC CURVES: A PROGRAMME

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ABSTRACT. Modular Koszul duality for chiral algebras is E_1 - E_1 operadic Koszul duality transported into the homotopical modular chiral realm on algebraic curves. This first section of the programme summary establishes the bar construction on an algebraic curve: the logarithmic propagator $\eta_{12} = d \log(z_1 - z_2)$, the Arnold relation on three points, and the ordered bar complex $\overline{B}^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$ with deconcatenation coproduct, together with the proof that $d^2 = 0$ at genus zero through cancellation of Arnold with the Borchers identity. The symmetric bar $\overline{B}^\Sigma(\mathcal{A})$ appears as the Σ_n -coinvariant shadow obtained by the lossy averaging map; the ordered bar is the primitive object and all five theorems of the programme extract Σ_n -invariant content from it.

I. THE BAR CONSTRUCTION ON ALGEBRAIC CURVES

The bar construction of an associative algebra is classical: given an augmented algebra $\mathcal{A}$ over a field, form the tensor coalgebra $T^c(s^{-1}\tilde{\mathcal{A}}) = \bigoplus_{n \geq 1} (s^{-1}\tilde{\mathcal{A}})^{\otimes n}$ on desuspended generators, equip it with the differential that encodes the multiplication, and observe that $d^2 = 0$ is equivalent to associativity. Koszul duality, homological algebra, and deformation theory all descend from this single construction.

A chiral algebra on a smooth algebraic curve X encodes operator product singularities: the product of two sections $a(z)b(w)$ diverges as $z \rightarrow w$, and the divergence carries algebraic data. The problem is to build the bar construction in this setting. The point is replaced by a curve; the multiplication map is replaced by a chiral bracket with poles along the diagonal; the tensor coalgebra must account for the geometry of colliding points on X .

I.1. The propagator. On the configuration space $C_2(X) = X \times X \setminus \Delta$ of two distinct points on a smooth algebraic curve X over $\mathbf{C}$, define

$$\eta_{12} = d \log(z_1 - z_2) = \frac{dz_1 - dz_2}{z_1 - z_2}. \quad (1.1)$$

This 1-form has a simple pole along the diagonal $\Delta \subset X \times X$ with residue 1. The residue is intrinsic: it depends on no coordinate, no metric, no level.

Three properties single out η_{12} . A double pole $dz/(z-w)^2$ transforms under coordinate change and has no coordinate-independent residue. A regular 1-form has residue zero and extracts nothing from the collision. The logarithmic derivative is the unique 1-form on $C_2(X)$ with a first-order pole along Δ and conformally invariant residue.

On $C_3(X)$, the three pullbacks $\eta_{12}, \eta_{23}, \eta_{31}$ satisfy

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0. \quad (1.2)$$

This is the *Arnold relation*. It restates the partial-fractions identity

$$\frac{1}{(z_1 - z_2)(z_2 - z_3)} + \frac{1}{(z_2 - z_3)(z_3 - z_1)} + \frac{1}{(z_3 - z_1)(z_1 - z_2)} = 0,$$

which holds because the left side, viewed as a rational function of z_1 with z_2, z_3 fixed, has simple poles at z_2 and z_3 whose residues sum to zero. Arnold [1] proved that (1.2), replicated across all triples $\{i, j, k\} \subset \{1, \dots, n\}$, generates the complete ideal of relations in $H^*(\text{Conf}_n(\mathbf{C}), \mathbf{C})$.

Under a coordinate change $z \mapsto w(z)$, the propagator transforms as $\eta_{ij} \mapsto \eta_{ij} + d \log \frac{w(z_i) - w(z_j)}{z_i - z_j}$. Near the diagonal the correction becomes $d \log w'(z_i)$: the standard cocycle for $\text{Diff}(X)$. The propagator is a BRST ghost for diffeomorphisms; its failure to be coordinate-invariant is the failure that the Virasoro algebra measures.

1.2. Chiral algebras. A *chiral algebra* $\mathcal{A}$ on X is a $\mathcal{D}_X$ -module equipped with a chiral bracket

$$\mu^{\text{ch}} : j_* j^* (\mathcal{A} \boxtimes \mathcal{A}) \longrightarrow \Delta_! (\mathcal{A}),$$

where $j : X \times X \setminus \Delta \hookrightarrow X \times X$ is the open complement and $\Delta_! \mathcal{A}$ is the $!$ -pushforward along the diagonal embedding [2]. The source consists of sections of $\mathcal{A} \boxtimes \mathcal{A}$ with poles of arbitrary order along Δ ; the chiral bracket extracts the singular part and projects it onto the diagonal. In local coordinates, the chiral bracket encodes the operator product expansion:

$$a(z) b(w) \sim \sum_{n \geq 0} \frac{a_{(n)} b(w)}{(z - w)^{n+1}}.$$

The n -th products $a_{(n)} b \in \mathcal{A}$ are the Fourier modes, extracted by $a_{(n)} b = \text{Res}_{z=w} [a(z) b(w) (z-w)^n]$. The zeroth product $a_{(0)} b$ is the Lie bracket (first-order pole); $a_{(1)} b$ carries the symmetric/metric data (second-order pole). Higher products $a_{(n)} b$ for $n \geq 2$ are determined by the $\mathcal{D}_X$ -module structure.

The Borcherds identity

$$\sum_{j \geq 0} \binom{m}{j} (a_{(n+j)} b)_{(m+k-j)c} = \sum_{j \geq 0} (-1)^j \binom{n}{j} \left(a_{(m+n-j)} (b_{(k+j)c}) - (-1)^n b_{(n+k-j)} (a_{(m+j)c}) \right) \quad (1.3)$$

is the complete axiom system. At $n = 0$ it reduces to the Jacobi identity for the Lie bracket $a_{(0)}$; at $n = 1$, $m = k = 0$, to the derivation property of $a_{(0)}$ with respect to $a_{(1)} b$.

1.3. The bar complex. Place sections $a_1, \dots, a_n \in \mathcal{A}$ at distinct points $z_1, \dots, z_n \in X$. The Fulton–MacPherson compactification $\overline{\text{FM}}_n(X)$ resolves $\text{Conf}_n(X)$ by iterated real oriented blowup along all diagonal strata [6]. Points of $\overline{\text{FM}}_n(X) \setminus \text{Conf}_n(X)$ record the limiting direction and relative scale of approach as points collide. The boundary is a manifold with corners; codimension-one faces D_S are indexed by subsets $S \subseteq \{1, \dots, n\}$ with $|S| \geq 2$, recording which points have collided. Each face factors:

$$D_S \simeq \overline{\text{FM}}_{|S|} \times \overline{\text{FM}}_{n-|S|+1},$$

cluster times residual.

The *ordered bar complex* is

$$\overline{B}_X^{\text{ord}}(\mathcal{A}) = \bigoplus_{n \geq 1} (s^{-1} \bar{\mathcal{A}})^{\otimes n} \otimes \Gamma(\overline{\text{FM}}_n(X), \Omega_{\log}^{n-1}),$$

with no Σ_n -quotient. Here $\bar{\mathcal{A}} = \ker(\varepsilon : \mathcal{A} \rightarrow \omega_X)$ is the augmentation ideal, s^{-1} denotes desuspension ($|s^{-1} a| = |a| - 1$), and the cohomological convention $|d| = +1$ is used throughout.

The bar differential decomposes:

$$d_{\bar{B}} = d_{\text{int}} + d_{\text{res}} + d_{\text{dR}}.$$

The internal differential d_{int} acts on each tensor factor; the de Rham differential d_{dR} acts on logarithmic forms on $\overline{\text{FM}}_n(X)$; and the residue differential d_{res} extracts OPE data at collision divisors. For a codimension-one face $D_{\{i,j\}}$ corresponding to the collision $z_i \rightarrow z_j$, the residue differential extracts the

p -th Fourier mode $a_{(p)}b$ of the OPE, where p is the pole order read from the form-theoretic coefficient of η_{ij} . The bar differential does not merely detect singularities; it extracts them, mode by mode, through the logarithmic kernel (1.1).

1.4. **Why $d^2 = 0$.** Expand $(d_{\text{int}} + d_{\text{res}} + d_{\text{dR}})^2$ into nine terms. Three vanish individually: $d_{\text{int}}^2 = 0$ (the underlying cochain complex), $d_{\text{dR}}^2 = 0$ (de Rham), and $d_{\text{int}} d_{\text{dR}} + d_{\text{dR}} d_{\text{int}} = 0$ (Leibniz). The remaining terms cancel pairwise, governed by index overlap:

- (a) *Disjoint supports.* When $\{i, j\} \cap \{k, l\} = \emptyset$, the faces $D_{\{i, j\}}$ and $D_{\{k, l\}}$ are transverse in $\overline{\text{FM}}_n(X)$. The residue operators commute: $d_{\text{res}}^{(ij)} d_{\text{res}}^{(kl)} + d_{\text{res}}^{(kl)} d_{\text{res}}^{(ij)} = 0$.
- (b) *Shared index.* When $\{i, j\} \cap \{k, l\} = \{j = k\}$, the triple collision $z_i = z_j = z_l$ sits in a codimension-two stratum. The form-theoretic contribution involves $\eta_{ij} \wedge \eta_{jl}$. The Arnold relation (1.2) identifies this with a cyclic sum over the three ordered approaches to the triple-collision point. On the algebraic side, the three terms produce $(a_{i(m)} a_j)_{(n)} a_l$, $(a_{j(m)} a_l)_{(n)} a_i$, $(a_{l(m)} a_i)_{(n)} a_j$ with appropriate pole orders, whose sum vanishes by the Borchers identity (1.3). The Arnold relation kills the form; the Borchers identity kills the algebra. Both are needed, and both are consumed.
- (c) *Same pair.* $d_{\text{res}}^{(ij)} d_{\text{res}}^{(ij)} = 0$ by the sign rule on desuspensions: extracting a residue twice at the same diagonal vanishes for degree reasons.

The result: $d_B^2 = 0$ if and only if the n -th products satisfy (1.3). The bar construction accepts exactly chiral algebras.

1.5. **Two coproducts.** The ordered bar $\overline{B}^{\text{ord}}(\mathcal{A})$ carries the *deconcatenation coproduct*:

$$\Delta[s^{-1}a_1 | \cdots | s^{-1}a_n] = \sum_{i=0}^n [s^{-1}a_1 | \cdots | s^{-1}a_i] \otimes [s^{-1}a_{i+1} | \cdots | s^{-1}a_n],$$

splitting an ordered sequence at every consecutive position: $n + 1$ terms. This is coassociative but not cocommutative. It makes $\overline{B}^{\text{ord}}(\mathcal{A})$ a cofree tensor coalgebra $T^c(s^{-1}\tilde{\mathcal{A}})$; the E_1 -coalgebra reflecting the linear ordering of points.

The Σ_n -coinvariant quotient

$$\overline{B}_X^\Sigma(\mathcal{A}) = \bigoplus_{n \geq 1} (s^{-1}\tilde{\mathcal{A}})_{\Sigma_n}^{\otimes n} \otimes \Gamma(\overline{\text{FM}}_n(X), \Omega_{\log}^{n-1})$$

carries the *coshuffle coproduct*, summing over all 2^n bipartitions of an unordered set: coassociative and cocommutative. This is the factorization coalgebra of Beilinson–Drinfeld on $\text{Ran}(X)$ [2], the E_∞ -coalgebra.

The ordered bar is the natural primitive. The symmetric bar is a quotient that loses data: the non-cocommutative structure of the ordered bar carries the spectral R -matrix, the Knizhnik–Zamolodchikov associator, and the full quantum group deformation, none of which survive the passage to Σ_n -coinvariants.

Remark 1.1 (Six bar complexes). Six bar-type objects coexist. The Francis–Gaitsgory bar [5] $B^{\text{Lie}}(\mathcal{A})$ is the cofree coLie coalgebra $\text{Lie}^c(s^{-1}\tilde{\mathcal{A}})$, capturing zeroth-pole data; it embeds into $\overline{B}^\Sigma$ via the inclusion $\text{Lie}^c \hookrightarrow \text{Sym}^c$ (a quasi-isomorphism in characteristic zero, but a categorical distinction). The genus-graded bar $B^{(g)}(\mathcal{A})$ extends $\overline{B}^\Sigma$ to families of curves over $\overline{\mathcal{M}}_g$, where the differential acquires curvature $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ (see §1.8). The modular bar $B^{\text{mod}}(\mathcal{A})$ restores $D^2 = 0$ by completing over the Feynman transform of the commutative modular operad. The E_1 -modular bar $B^{E_1\text{-mod}}(\mathcal{A})$, an algebra over the

Feynman transform of the associative modular operad, is the primitive among these; the modular bar is its Σ_n -coinvariant image. No two of these six objects are isomorphic in general.

1.6. Three operations on the bar. The bar construction is a categorical logarithm: $B(\mathcal{A} \otimes \mathcal{A}') \simeq B(\mathcal{A}) \oplus B(\mathcal{A}')$. The logarithmic form $\eta_{ij} = d \log(z_i - z_j)$ is the integral kernel; the Arnold relation is the cocycle condition for log.

Three distinct functors act on $B(\mathcal{A})$ and produce three distinct outputs:

$$\begin{aligned} \Omega(B(\mathcal{A})) &\simeq \mathcal{A} && \text{(cobar inversion: recovers the original),} \\ \mathbb{D}_{\text{Ran}}(B(\mathcal{A})) &\simeq \mathcal{A}_\infty^! && \text{(Verdier duality: the homotopy Koszul dual),} \\ (H^*(B(\mathcal{A})))^\vee &= \mathcal{A}^! && \text{(linear duality of bar cohomology: the strict dual).} \end{aligned} \tag{1.4}$$

Cobar inversion recovers the original algebra $\mathcal{A}$ itself; it is a round-trip, not a duality. Verdier duality on $\text{Ran}(X)$ converts the bar coalgebra into a factorization algebra $\mathcal{A}_\infty^!$, the homotopy Koszul dual retaining full chain-level data. That its cohomology recovers the strict Koszul dual $\mathcal{A}^!$ on the Koszul locus is the content of the Verdier intertwining theorem, not a tautology. Confusing any two of (1.4) is a category error.

1.7. The Heisenberg algebra at bar degree two. The Heisenberg chiral algebra $\mathcal{H}_k$ is generated by a single field α of conformal weight 1 with OPE

$$\alpha(z) \alpha(w) \sim \frac{k}{(z-w)^2}.$$

Only the second-order pole is present: $\alpha_{(1)} \alpha = k \cdot \mathbf{1}$ and $\alpha_{(n)} \alpha = 0$ for $n \neq 1$. No first-order pole: $\alpha_{(0)} \alpha = 0$, no Lie bracket. All triple products vanish since $\mathbf{1}_{(n)} = 0$ for $n \geq 0$.

The bar complex at tensor degree 2 gives

$$d_{\text{res}}(s^{-1} \alpha \otimes s^{-1} \alpha \otimes \eta_{12}) = \alpha_{(1)} \alpha = k \cdot \mathbf{1}.$$

The level k is visible already at genus 0. At tensor degree $n \geq 3$: every residue at a collision stratum $D_{\{i,j\}}$ produces $\alpha_{(1)} \alpha = k \cdot \mathbf{1}$ on the pair (i, j) ; since $\mathbf{1}_{(p)} \alpha = 0$ for $p \geq 0$, no further contractions occur. All structure is captured at tensor degree 2: the bar complex is quadratic. The Heisenberg algebra is chirally Koszul.

The ordered bar carries the collision residue

$$r(z) = \frac{k}{z}.$$

The second-order OPE pole $k/(z-w)^2$ drops by one order under the logarithmic kernel $d \log(z-w)$: the bar differential absorbs one power of $(z-w)$, producing the first-order r -matrix pole. Applying the averaging map:

$$\text{av}(r(z)) = k = \kappa(\mathcal{H}_k).$$

For the Heisenberg algebra, the R -matrix is the modular characteristic: $r(z) = k/z$ is scalar, so averaging loses nothing.

The deficiency: no first-order pole, no Lie bracket, no matrix structure in the collision residue, no quantum group. Everything that makes representation theory nontrivial is absent. The simplest algebra where the R -matrix is matrix-valued is the first algebra with a first-order pole: the affine Kac–Moody algebra.

Example 1.2 (Kac–Moody collision residue). The affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ at level k is generated by currents J^a , $a = 1, \dots, \dim \mathfrak{g}$, with OPE

$$J^a(z) J^b(w) \sim \frac{f^{ab}{}_c J^c(w)}{z-w} + \frac{k \kappa^{ab}}{(z-w)^2}.$$

Both poles are present. The ordered bar carries the collision residue

$$r(z) = \frac{k \Omega}{z}, \quad \Omega = \sum_a J^a \otimes J_a \in \mathfrak{g} \otimes \mathfrak{g},$$

where Ω is the Casimir tensor and k is the level. At $k = 0$ the r -matrix vanishes: no level, no collision data. The averaging map collapses $k \Omega$ to a scalar:

$$\text{av}(k \Omega / z) = \frac{k \dim(\mathfrak{g})}{2h^\vee} = \kappa_{\text{dp}}(\widehat{\mathfrak{g}}_k),$$

where $h^\vee$ is the dual Coxeter number. The full modular characteristic adds the Sugawara shift:

$$\kappa(\widehat{\mathfrak{g}}_k) = \kappa_{\text{dp}}(\widehat{\mathfrak{g}}_k) + \frac{\dim(\mathfrak{g})}{2} = \frac{\dim(\mathfrak{g}) (k + h^\vee)}{2h^\vee}.$$

The matrix structure of Ω , erased by Σ_2 -coinvariance, encodes the representation theory of $\mathfrak{g}$: it is the datum that builds the Yangian $Y(\mathfrak{g})$.

1.8. Genus 0 and genus ≥ 1 . Everything above takes place at genus 0: the propagator $\eta_{12} = d \log(z_1 - z_2)$ is globally defined on $\mathbb{P}^1$, and the Arnold relation holds without correction. On a curve of genus $g \geq 1$, the function $z_1 - z_2$ has no global meaning: the curve is compact, and any meromorphic function with a single simple zero must also have a simple pole.

The replacement is the *prime form* $E(z_1, z_2)$, a section of $K^{-1/2} \boxtimes K^{-1/2}$ vanishing to first order along the diagonal with no other zeros. The genus- g propagator

$$\eta_{12}^{(g)} = d \log E(z_1, z_2) + (\text{period correction})$$

is single-valued but no longer holomorphic; the period correction couples to the Hodge bundle $\mathbb{E} \rightarrow \overline{\mathcal{M}}_g$ through the period matrix. The propagator $d \log E(z_1, z_2)$ has conformal weight 1 in both variables, regardless of the conformal weight of the field being sewed: all edges of the genus- g graph sum use the standard Hodge bundle $\mathbb{E}_1 = R^0 \pi_* \omega_\pi$.

The Arnold relation acquires a defect, and the fibrewise bar differential satisfies

$$d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g,$$

where ω_g is the Arakelov form on the fibre of $\pi: C_g \rightarrow \overline{\mathcal{M}}_g$ and $\kappa(\mathcal{A}) \in \mathbf{C}$ is the *modular characteristic*, determined entirely by the leading OPE singularity. Explicit values:

$$\kappa(\mathcal{H}_k) = k, \quad \kappa(\widehat{\mathfrak{g}}_k) = \frac{\dim(\mathfrak{g}) \cdot (k + h^\vee)}{2h^\vee}, \quad \kappa(\text{Vir}_c) = \frac{c}{2}.$$

Period integrals restore nilpotence: the Gauss–Manin connection absorbs the fibrewise curvature, and the total differential satisfies $D_g^2 = 0$.

Genus 0 is uncurved algebra. Genus ≥ 1 is curved geometry. The entire genus tower is controlled by a single scalar: the curvature of the bar complex on a family of curves.

Remark 1.3 (Recovering the classical bar). When $X = \text{Spec } \mathbf{C}$ is a point, configuration spaces collapse, logarithmic forms vanish, and the chiral bar complex recovers the classical bar construction of an associative algebra $\mathcal{A}$. This recovery requires specifying the retraction data: vertex algebras live on the formal disk D , and the passage from D to a bare point discards the thickening (completion, growth

conditions) on which the $\mathcal{D}$ -module structure depends. A quadratic algebra $A = T(V)/(R)$ has quadratic dual $A^! = T(V^*)/(R^\perp)$; when A is Koszul, the bar cohomology concentrates in diagonal degrees and $H^*(B(A)) \cong (A^!)^\vee$. The operadic instances $\text{Com}^! = \text{Lie}$ and $\text{Sym}^! = \Lambda$ recover the Chevalley–Eilenberg complex as a restricted bar construction.

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